

1. Regression

1.1. Simple Linear Regression

$$\begin{cases} S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 \\ S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \Rightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \\ S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 \end{cases}$$

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2/S_{xx}}} \sim \mathcal{N}(0, 1), \frac{\hat{\beta}_0 - \beta_0}{\sqrt{(\sigma^2 \sum_{i=1}^n x_i^2)/(nS_{xx})}} \sim \mathcal{N}(0, 1)$$

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{s^2/S_{xx}}} \sim t_{n-2}, \frac{\hat{\beta}_0 - \beta_0}{\sqrt{(s^2 \sum_{i=1}^n x_i^2)/(nS_{xx})}} \sim t_{n-2}$$

$$\frac{\hat{y}_0 - \mathbb{E}[y_0]}{s\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} \sim t_{n-2}, \frac{y_{\text{new}} - \hat{y}_{\text{new}}}{s\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} \sim t_{n-2}$$

$$\mathcal{R}^2 = \frac{\text{SSR}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = r_{xy}^2$$

$\mathcal{R}^2$ of the observed variation in the response variable can be explained by the regression model.

$$\mathbb{E}[\text{MS}_{\text{Reg}}] = \sigma^2 + \beta_1 S_{xx}$$

$$\hat{\beta}_0 = l^\top y, \hat{\beta}_1 = k^\top y, k_i = \frac{x_i - \bar{x}}{S_{xx}}, l_i = \frac{1}{n} - k_i \bar{x}$$

1.2. Multiple Linear Regression

1.2.1. Useful Formula

$$\mathbb{E}[yy^\top] = \mathbb{E}[y]\mathbb{E}[y]^\top + \Sigma_y$$

$$\mathbb{E}[y^\top Ay] = \mathbb{E}[y]^\top A \mathbb{E}[y] + \text{tr}(A \Sigma_y)$$

$$\text{Cov}(Ax + b) = A \text{Cov}(x) A^\top$$

$$x \perp y \Rightarrow \Sigma_{x+y} = \Sigma_x + \Sigma_y$$

$$\text{Outliers: } |e_i^*| = \left| \frac{e_i}{\sqrt{\text{MSE}}} \right| > 3$$

QQ-Plot: 左边：往下尾巴大；右边：往上尾巴大

1.2.2. Normal Distributions

$$\frac{1}{(2\pi)^n |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(y - \mu)^\top \Sigma^{-1}(y - \mu)\right)$$

$$a^\top y \sim \mathcal{N}(a^\top \mu, a^\top \Sigma a)$$

1.2.3. Definition

$$y = X\beta + \varepsilon, \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

$$y, \beta, \varepsilon \in \mathbb{R}^{n \times 1}, X \in \mathbb{R}^{n \times (p+1)}$$

$$\hat{\beta} = (X^\top X)^{-1} X^\top y, e = y - X\hat{\beta}$$

1.2.4. Hat Matrix

$$H = X(X^\top X)^{-1} X^\top \in \mathbb{R}^{n \times n}$$

$$\hat{y} = Hy, e = (I - H)y$$

H 和 $I - H$ 是投影矩阵, $Hy \perp (I - H)y$:

$$\begin{cases} HX = X \\ H^2 = H \\ (I - H)^2 = I - H \end{cases}$$

H 和 $I - H$ are orthogonal matrices, so

$$\begin{cases} H^\top = H \\ (I - H)^\top = I - H \end{cases}$$

1.2.5. Sum of Squares

$$A \leftarrow \frac{1}{n} 11^\top \quad (\text{全 } \frac{1}{n} \text{ 矩阵, } \mathbb{R}^{n \times n})$$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = y^\top (I - A)y$$

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = e^\top e = y^\top (I - H)y$$

$$\text{SSR} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = y^\top (H - A)y$$

$$\text{SSR} \sim \sigma^2 \chi_p^2, \text{SSE} \sim \sigma^2 \chi_{n-p-1}^2$$

1.2.6. ANOVA Table

Source	DF	SS	MS	$\mathcal{F}$
Regression	p	SSR	MS_{Reg}	F
Residual	$n - p - 1$	SSE	s^2	
Total	$n - 1$	S_{yy}		

$$\mathbb{E}[s^2] = \sigma^2, F = \frac{\text{MS}_{\text{Reg}}}{s^2} \sim \mathcal{F}(p, n - p - 1)$$

$$\mathcal{H}_0: \beta_1 = \dots = \beta_p = 0 \Rightarrow p = \mathbb{P}(\mathcal{F}_{p, n-p-1} > F)$$

1.2.7. Cochran's Theorem

$y \sim \mathcal{N}_n(0, \sigma^2 I)$, A_i 满足:

- $\sum_{i=1}^m A_i = I$
- $A_i^2 = A_i$ (投影矩阵)
- $A_i^\top = A_i$

我们有:

- $yA_i y$ 是互相 independent 的
- $yA_i y \sim \sigma^2 \chi_{r_i}^2$, 其中 $r_i = \text{tr}(A_i)$

1.2.8. $\mathcal{R}^2$ Statistic

$$\mathcal{R}^2 = \frac{\text{SSR}}{S_{yy}}$$

$p \uparrow$ 会导致 $\mathcal{R}$ 接近 1, 因此我们引入 adjusted $\mathcal{R}^2$:

$$\mathcal{R}_a^2 = 1 - \frac{\text{SSE}/(n-p-1)}{S_{yy}/(n-1)}$$

$\mathcal{R}^2$ too small $\rightarrow$ cannot predict precisely

$\mathcal{R}^2$ too large $\rightarrow$ overfit

1.2.9. t Statistic

$$\hat{\beta} \sim \mathcal{N}_{p+1}\left(\beta, \sigma^2 (X^\top X)^{-1}\right)$$

$$\hat{\beta}_i \sim \mathcal{N}\left(\beta_i, \sigma^2 (X^\top X)^{-1}_{i+1, i+1}\right)$$

$$\mathcal{H}_0: \beta_i = 0 \Rightarrow t^* = \frac{\hat{\beta}_i}{\text{SE}(\hat{\beta}_i)} = \frac{\hat{\beta}_i}{\sqrt{s^2 (X^\top X)^{-1}_{i+1, i+1}}}$$

1.2.10. Extra Sum of Squares

$$\begin{aligned} \text{SSR}(X_2 | X_1) &= \text{SSR}(X_1, X_2) - \text{SSR}(X_1) \\ &= \text{SSE}(X_1) - \text{SSE}(X_1, X_2) \end{aligned}$$

$$\text{SSEXT} = \text{SSE}_\omega - \text{SSE}_\Omega = \text{SSR}_\Omega - \text{SSR}_\omega$$

Source	DF	SS	MS	$\mathcal{F}$
ω	q	SSR_ω	$\frac{\text{SSR}_\omega}{q}$	
Extra	$p - q$	SSEXT	$\frac{\text{SSEXT}}{p - q}$	F
Residual	$n - p - 1$	SSE	$\frac{\text{SSE}_\Omega}{n - p - 1}$	
Total	$n - 1$	S_{yy}		

$$F = \frac{\text{SSEXT}/(p-q)}{\text{SSE}_\Omega/(n-p-1)} \sim \mathcal{F}(p - q, n - p - 1)$$

$\mathcal{H}_0$: We should include the extra variables

$$p = \mathbb{P}(\mathcal{F}_{p-q, n-p-1} > F).$$

2. Classification

2.1. One-Way Classification Model

2.1.1. Definition

- Factor:** Gender | Race
- Levels:** M, F | CN, JA, RU
- No. of levels:** 2 | 3
- Treatments:** (M, CN), (F, CN), ...
- No. of treatments:** $2 \times 3 = 6$

$$y_{ij} = \theta_i + \varepsilon_{ij}, \quad i \in [1, k], j \in [1, n_i], \quad \varepsilon \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

Re-parameterization: $\theta_i = \mu + \tau_i$; $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$

- Overall sample sum: $y_{..} = \sum_{i=1}^k \sum_{j=1}^{n_i} y_{ij}$
- i -th sample sum: $y_{i.} = \sum_{j=1}^{n_i} y_{ij}$
- Overall sample mean: $\bar{y}_{..} = \frac{y_{..}}{n}$
- i -th sample mean: $\bar{y}_{i.} = \frac{y_{i.}}{n_i}$

2.1.2. Sum of Squares

$$\text{Correction factor: c.f.} = n(\bar{y}_{..})^2 = \frac{y_{..}^2}{n}$$

$$\begin{aligned} \text{Total corrected SS: } S_{yy} &= \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2 = \\ &= \sum_{i=1}^k \sum_{j=1}^{n_i} y_{ij}^2 - \text{c.f.} \end{aligned}$$

$$\text{Between-group SS: SST} = \sum_{i=1}^k n_i (\bar{y}_{i.} - \bar{y}_{..})^2 = \sum_{i=1}^k \frac{y_{i.}^2}{n} - \text{c.f.}$$

Within-group SS: $SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2 = S_{yy} - SST$

2.1.3. One-way ANOVA Table

Source	DF	SS	MS	$\mathcal{F}$
Between groups	$k - 1$	SST	MST	F
Within groups	$n - k$	SSE	MSE	
Total	$n - 1$	S_{yy}		

$MST = \frac{SST}{k}, MSE = \frac{SSE}{n-k}, \mathbb{E}[MSE] = \sigma^2$

$F = \frac{MST}{MSE} \sim \mathcal{F}_{k-1, n-k}$

2.1.4. Analysis of Parameters

我们很多时候喜欢研究关于 θ 的一些线性式子，比如比较两个类别 $(\theta_i - \theta_j)$ 等等，一般地：

$L = \alpha^\top \theta, \hat{L} = \alpha^\top \bar{y}$

$\hat{L} \sim \mathcal{N}\left(L, \sigma^2 \sum_{i=1}^k \frac{\alpha_i^2}{n_i}\right)$

$\hat{L} \sim t_{n-k}\left(L, s^2 \sum_{i=1}^k \frac{\alpha_i^2}{n_i}\right)$

2.2. Two-Way ANOVA

2.2.1. Definition

$y_{ijk} = \theta_{ij} + \varepsilon_{ijk}, \varepsilon_{ijk} \sim \mathcal{N}(0, \sigma^2)$

$i = 1, ..., I, j = 1, ..., J, k = 1, ..., r$

2.2.2. Re-parameterization

$\theta_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij}$

$\sum \alpha_i = \sum \beta_j = \sum_{i=1}^I (\alpha\beta)_{ij} = \sum_{j=1}^J (\alpha\beta)_{ij} = 0$

2.2.3. Model Fitting

$$\begin{cases} \hat{\mu} = \bar{y}_{...} \\ \hat{\alpha}_i = \bar{y}_{i..} - \hat{\mu} \\ \hat{\beta}_j = \bar{y}_{.j.} - \hat{\mu} \\ \hat{\alpha\beta}_{ij} = \bar{y}_{ij.} - \hat{\mu} - \hat{\alpha}_i - \hat{\beta}_j \end{cases}$$

2.2.4. Sum of Squares

Correction factor: c.f. = $\frac{y_{..}^2}{IJr}$

$S_{yy} = \sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{...})^2$

$SST = r \sum_i \sum_j (\bar{y}_{ij.} - \bar{y}_{...})^2$

$SSE = \sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{ij.})^2$

$SS_A = rJ \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2$

$SS_B = rI \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2$

$SS_{AB} = r \sum_i \sum_j (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2$

$S_{yy} = SST + SSE$

2.2.5. Two-way ANOVA Table

Source	DF	SS	MS	$\mathcal{F}$
Main eff. A	$I - 1$	SS_A	MS_A	F_A
Main eff. B	$J - 1$	SS_B	MS_B	F_B
Interaction	$(I - 1)(J - 1)$	SS_{AB}	MS_{AB}	F_{AB}
Error	$IJ(r - 1)$	SSE	MSE	
Total	$IJr - 1$	S_{yy}		

$MS_A = \frac{SS_A}{I-1}, MS_B = \frac{SS_B}{J-1}, MS_{AB} = \frac{SS_{AB}}{(I-1)(J-1)}$

$\mathbb{E}[MS_A] = \sigma^2 + rJ \sum_i \frac{(\bar{\theta}_{i.} - \mu)^2}{I-1}$

$\mathbb{E}[MS_B] = \sigma^2 + rI \sum_j \frac{(\bar{\theta}_{.j} - \mu)^2}{J-1}$

$\mathbb{E}[MS_{AB}] = \sigma^2 + r \sum_i \sum_j \frac{(\theta_{ij} - \bar{\theta}_{i.} - \bar{\theta}_{.j} + \mu)^2}{(I-1)(J-1)}$

$MSE = \frac{SSE}{IJ(r-1)}, \mathbb{E}[MSE] = \sigma^2$

2.2.6. Analysis of Parameters

Parameters of interest: $L = \sum_{i=1}^I \sum_{j=1}^J \alpha_{ij} \theta_{ij}$

$\hat{L} = \sum_{i=1}^I \sum_{j=1}^J \alpha_{ij} \bar{y}_{ij}$

$\hat{L} \sim \mathcal{N}\left(L, \sigma^2 \sum_{i=1}^I \sum_{j=1}^J \frac{\alpha_{ij}^2}{r}\right)$

$\hat{L} \sim t_{IJ(r-1)}\left(L, s^2 \sum_{i=1}^I \sum_{j=1}^J \frac{\alpha_{ij}^2}{r}\right)$

3. Universal Linear Modelling

3.1. Dummy Variable

Similar to one-hot: k levels $\Rightarrow I_1, I_2, ..., I_{k-1}$

3.2. ANCOVA

3.2.1. One Factor

$Y = \mu + \tau_1 T_1 + \dots + \tau_{k-1} T_{k-1} + \gamma_1 X_1 + \dots + \gamma_p X_p + \varepsilon$

Source	DF	SS	MS	$\mathcal{F}$
Reg. fitting	p	SSR_ω	MSR_ω	
Extra	$k - 1$	SSEXT	MSEXT	F
Residual	$n - p - k$	SSE_Ω	MSE_Ω	
Total	$n - 1$	S_{yy}		

$MSR_\omega = \frac{SSR_\omega}{p}, MSEXT = \frac{SSEXT}{k-1}, MSE_\Omega = \frac{SSE_\Omega}{n-p-k}$

$F = \frac{MSEXT}{MSE_\Omega} \sim \mathcal{F}(k - 1, n - p - k)$

3.2.2. Two Factors

$$Y = \mu + \sum_{i=1}^{I-1} \alpha_i A_i + \sum_{j=1}^{J-1} \beta_j B_j + \sum_{i=1}^{I-1} \sum_{j=1}^{J-1} (\alpha\beta)_{ij} A_i B_j + \sum_{k=1}^p \gamma_p X_p + \varepsilon$$

Reduce both:

Source	DF	SS	MS	$\mathcal{F}$
Reg. fitting	$p + I + J - 2$	SSR_ω	MSR_ω	
Extra	$(I - 1)(J - 1)$	SSEXT	MSEXT	F
Residual	$n - p - IJ$	SSE_Ω	MSE_Ω	
Total	$n - 1$	S_{yy}		

$MSR_\omega = \frac{SSR_\omega}{p+I+J-2}, MSEXT = \frac{SSEXT}{(I-1)(J-1)}, MSE_\Omega = \frac{SSE_\Omega}{n-p-IJ}$

$F = \frac{MSEXT}{MSE_\Omega} \sim \mathcal{F}((I - 1)(J - 1), n - p - IJ)$

Reduct one:

Source	DF	SS	MS	$\mathcal{F}$
Concomitant	p	SSR''		
Main eff. B	$J - 1$	SS_B	MS_B	F_B
Main eff. A	$I - 1$	SS_A	MS_A	F_A
Interaction	$(I - 1)(J - 1)$	SSEXT	MSEXT	F_{AB}
Error	$n - p - IJ$	SSE_Ω	MSE_Ω	
Total	$n - 1$	S_{yy}		

$F_A = \frac{MS_A}{MSE_\Omega}, F_B = \frac{MS_B}{MSE_\Omega}, F_{AB} = \frac{MSEXT}{MSE_\Omega}$

Interaction between factors and concomitant variables:

$$Y = \mu + \sum_{i=1}^{I-1} \alpha_i A_i + \sum_{j=1}^{J-1} \beta_j B_j + \sum_{i=1}^{I-1} \sum_{j=1}^{J-1} (\alpha\beta)_{ij} A_i B_j + \left[\sum_{i=1}^{I-1} \delta_i A_i + \sum_{j=1}^{J-1} \zeta_j B_j + \sum_{i=1}^{I-1} \sum_{j=1}^{J-1} (\delta\zeta)_{ij} A_i B_j \right] X + \gamma X + \varepsilon$$